

ActInf GuestStream #016.1 ~ Mark Solms "Consciousness as Precision Op

GuestStream | Mark Solms | 1:57:50

All right. Hello, everyone. Welcome to Acton Flab guest stream number 16.1 with Mark Solms. It's January 22, 2022. It's going to be an awesome discussion. We really appreciate everyone taking the time out to participate in the live stream and in the discussion. We're going to have some intermittent presentation intervals by Professor Solms and also a period for discussion alternating with that. So we'll each just introduce ourselves and then pass it directly to the presentation section. So I'm Daniel. I'm a researcher in California and I'll pass to Dave. I'm Dave. I'm a software developer and student of psychology. I live 200 kilometers north of Manila. Stephen.

Hello. I'm Stephen. I'm based in Toronto, Canada, and I'm doing a practice-based PhD around social topographies and how people engage in community development. And I'll pass it over to Mark now. Welcome.

Thanks very much. So what I propose to do is to, in a moment, I'll share my screen and you'll see my slides. I'm going to start with the summary slide of all the points that I propose to cover. And then I'll take them one by one, each of those points. And at the end of each of them, I'll stop and see if anyone wants to comment or ask a question. And then we slowly plow through it like that. We've got two hours, so it should be okay. But if worst comes to worst and we don't get to the end of the list of topics, we will arrange a follow-up meeting to conclude them. So I'm very glad to be here. Thanks for inviting me. And I'm really looking forward to this discussion. Here come my slides. On the screen is the title of my book. And here is the promised summary of what I'm going to cover. As I said, I'll cover each of these points in turn. And then we'll have questions and comments in between. And if we don't get to the end, that's fine. Let's rather give each one of the topics the time that it needs. So I start with the assertion that affect is the foundational form of consciousness and that affect is intrinsically conscious. I then move to make the point that that claim, that affect is intrinsically conscious, in other words, that you can't have a feeling that you don't feel, that this is not based on semantic arguments, but rather on empirical evidence. And I will summarize the evidence. I'll then make the point that affect is not synonymous with interceptive inference, which is widely believed these days. And I'll tell you why that matters when I get there. And then I'll move on to making the point that all of this casts new light on the hard problem, particularly if affect is not just an interceptive form of perception, because there are lots of problems about perception that have got us into this hard problem in the first place. I'll argue that affect, rather than being interceptive inference, that it's an extended form of homeostasis. And I'll tell you what that, what I mean by that when we get there. And I'll make the point that complex organisms like ourselves require multiple homeostats, and that these are to be treated as categorical variables, which matters. And that these variables must be prioritized. One can't meet all one's needs simultaneously. I'll then, this is a large point. This one is a large point, the empirical evidence about

affect. And this one is another large point. The mechanism of perceptual and cognitive consciousness is precision modulation of allostatic prediction errors. That's a big claim. And so we must spend some time on it. I'll then talk about the nature of the predictive hierarchy. I seem to see it a little differently from most people. So I want to set out my stall on that score and explain why I think the predictive hierarchy is arranged concentrically. And that this, with the, as one moves toward the periphery of the hierarchy, there's progressively increasing tolerance of uncertainty. And then lastly, I'll say a word or two about why I think that consciousness can be artificially engineered.

So even if we don't get to the end of all of these topics, one way or another, even if we're the second meeting, those are the topics we will cover. And we'll see how far we get tonight. And at least, you know what the over, what the overall, when I say tonight, I'm in Berlin, I know it's morning for some of you. But at least you get the overall picture from that slide. So let's start with, affect is the foundational form of consciousness, and it's intrinsically conscious.

The discoverer of the unconscious, I quote him, because Freud more than anybody wanted us to realize just how much of mental life is unconscious. Even Freud said that it is surely of the essence of an emotion that we should be aware of it. That is, that it should become known to consciousness. Thus, the possibility of the attribute of unconsciousness would be completely excluded as far as emotions, feelings, and affects are concerned. It's quite a startling statement from, as I said, the discoverer of the unconscious to say that the possibility of the attribute of unconsciousness would be completely excluded as far as emotions, feelings, and affects are concerned. Now, I know this is a controversial issue, and that there are all sorts of ways in which people want to argue that there are such things as unconscious emotions and whatnot. And this has partly to do with the fact that different people mean different things by these words, emotions, affects, and so on. And it has, there are also all kinds of methodological operationalizations of these things.

I won't go into all of that. I'm happy to, if any of you want to discuss it. But I've, I've emphasized the word feelings to make clear that if anyone wants to argue that they think that the word emotion can apply to unconscious processes or affects can apply to unconscious processes, that's fine by me. But the word feelings, I don't see how you can ever argue that feelings can be unconscious. It just doesn't make any sense that a feeling could exist without being felt. If it were not felt, it would not be a feeling. And so I'm going to focus everything that I'm going to say on this process, this, this thing, this, this mental state called feelings. And I'm arguing that feelings are the foundational form of consciousness, and that these, this form of consciousness is intrinsically conscious. David Vogelpohl, MD, So that's a sort of a dogmatic statement. I'm not sure if anyone wants to discuss that. Perhaps you want to hold your horses. But that's the end of my first point. So anyone who wants to interrupt here, go ahead. Otherwise, I'll move on.

David Vogelpohl, MD, Thanks for the digestible claim based format. My question would be, what else has been posited in your view, as a foundational form of consciousness? So what are the counterfactuals for that first claim? What other horses are in the race?

Dr. Peter Van Doren, MD, The reason I focus on that fit.

Well, I shouldn't say the reason I focus on that because that's not quite what you're asking, you're asking, what are the other horses in the race? So let me say that the whole search for a neural correlative consciousness, which is in a way where the trouble began with Francis Crick, it was focused on visual perception.

And there are good reasons to focus on visual perception, mainly because if you take a look at your, you'll excuse the pun, take a look at your consciousness right now, it's probably dominated by visual perception. Visual qualia predominates in our consciousness. And a very large part of our cortex is given over to visual processing. And everybody's always believed that cortex is where it's at when it comes to consciousness. And we also understand a very great deal about visual cortical mechanisms and visual mechanisms altogether. So that was a perfectly reasonable place to start. So that's the other main horse that's in the race. But vision was thought to be a good model example, not because it is the totality of consciousness, but because it was assumed that if we can crack the problem of how visual consciousness arises, then we can generalize our conclusions to other forms. And I'm going to argue that there's something profoundly different about visual consciousness and affective consciousness. And moreover, that there's something profoundly different about cortical consciousness. And the brainstem mechanisms, which is where affective consciousness arises, and I know that

I'm saying that boldly now, and I'm about to argue that point in the second of the bullet points on the screen. Affective consciousness is not a cortical form of consciousness, it's a brainstem form. And that the cortical form of consciousness is contingent upon the brainstem form. And this is one of the main reasons, apart from evolutionary reasons, which sort of interdigitate with and flow from that, from that anatomical localization. Why I think that affect is the foundational form. So it's really vision that I'm using as my kind of foil to answer the question simply. Stephen.

Stephen Sackler

Yeah, one question with feelings is, would you say that feelings are integrated with proprioception, in the sense that they are felt sense of where the body is as well, so it brings it into kind of an action orientated domain? Or is it enough to have sort of a sense of how things are going, sort of an overview of just an emotional state? And then whether those two connect in some way?

Stephen Sackler

As I'm going to argue, when I get to my third point, I don't think that affect is synonymous with interoceptive inference altogether, interoceptive perception altogether. And I don't know, some people would call, would, would, would class proprioception with interoception. Some people would not, some people would say it's a sort of a, it's a sort of a hybrid between interoceptive and extraoceptive perception. But to the extent that by the word proprioception, you're referring to a form of perception. I would disagree with that, because I don't think that affect is a form of perception.

I think that proprioception follows, if by proprioception you mean what, what urologists normally mean by that word, it's, it's a, it's, it's part of the broader category of somatic sensation.

And therefore, it does not qualify, I think as affective. Because as I say, I'm going to argue

that affect is not a form of perception. But there, there, there, I draw a distinction, as do, you know, I think it's, it's perfectly fine to draw a distinction between extraoceptive bodily sensations and interoceptive bodily sensations. But I think that the interoceptive type, um, is, is, is, is nothing other than perception, um, and not intrinsically different. So even if you think of proprioception as straddling those two domains, I still think that it's a modality of perception, perception, and suffers from all the same problems as vision for that reason.

I see we have a comment in the chat line, I'm not sure how, how we're going to deal with that if one of you's, um, I'll read this, it was a question in the live chat from Ingrid, they wrote, what differentiates affect from feeling from your perspective? Is affect conscious or pre-conscious? Thank you.

You know, this is a, a, a, a perennial problem. Uh, we don't all mean the same thing by these terms. So that's why, uh, I've focused on the word feeling because there can be little doubt what feeling denotes. It denotes something you feel. In other words, something that you are, uh, necessarily aware of, if you were not aware of it, you, it wouldn't be called a feeling because you wouldn't feel it.

Um, affects and emotions denote many different things to different people. So, uh, when you ask me, is affect pre-conscious or conscious and so on, it depends what you mean by affect, um, to my mind, and I can only say, um, you know, just make clear how I use these words. Uh, I use the word feeling, uh, to denote the subjective experience of affect. Uh, affect is a kind of abstract term for a function, uh, that we feel, um, but that doesn't mean that you all have to use the word in the same way. And it's precisely in order to not get lost in this kind of semantic tangle, uh, that I'm focusing on the word feeling. Uh, and, uh, I'll make clear, uh, uh, in, in the very point that I'm moving to next, uh, what I mean by feeling in terms of it's, um, it's, it's physiological and anatomical and other objective, uh, um, uh, instantiations. Excellent. Steven, you, you have your hand up probably just did, or is that your previous hand? So just go to the second point. Okay. Very good. So the second point is, as you saw, we've just been talking about the meaning of the words. Um, and I've, uh, I've, um, staked my claim, uh, for the word feeling, uh, to make clear that that's what I mean when I say that affect is intrinsically conscious, I mean that it is necessarily felt. Um, I I'm also saying it's the foundational form of consciousness that doesn't flow from the word, uh, except perhaps from the word in the sense that the way that Nagel and Chalmers, uh, define consciousness and many people seem to have got sort of uh, agreed with this way of defining it, that there is something it is like to be conscious that, that this something it is likeness, uh, uh, it seems to be more or less the same thing as feeling. It seems to mean the same thing, but feeling in a rather broader sense of the word than affective feeling. Uh, so I want to be clear now, uh, why other than those, other than those linguistic or semantic points, uh, why I'm making this claim, uh, about feeling it's not by any means based in the semantics that you have to feel appealing. I, I want to be clear, uh, why I think, uh, that this is the foundational form of consciousness. We've, we've all known since 1948, uh, when there was this big surprise that Magoon and Maritzi found, uh, that all consciousness is contingent upon, um, what we subsequently called reticular activating arousal. In other words, upon the, something that's contributed by these core brainstem structures, uh, which after Magoon and Maritzi's work came to be known as the

reticular activating system. I want to be clear. I'm, uh, not only, uh, talking about the reticular activating system, but also about the peri-acaductal gray, which is not, uh, technically part of the reticular activating system, but it is part of, uh, the brain structures, uh, which seem to be fundamental, uh, to the generating of consciousness. Um, the, what Magoon and Maritzi showed among other things was that if you make a small lesion above the reticular activating system, thereby severing it, uh, disconnecting it from the forebrain and from the cortex, uh, in, in particular, uh, then consciousness, uh, is obliterated entirely. Cortical consciousness is 100% dependent upon brainstem arousal. Um, that was a shock, uh, and a shock that I don't think, uh, we have fully come to terms with, uh, even all these years later. Uh, it should have led us to realize that we've been looking in the wrong place, uh, for, uh, the neural correlative consciousness. If we're looking to cortex, because it's plain, even from just that study performed in 1948, uh, that cortical consciousness, uh, is derived from elsewhere. Cortical conscious consciousness cannot, uh, occur, uh, in the absence of reticular activating arousal. But, uh, the way that Magoon and Maritzi, uh, uh, fudged it was to say, and everyone more or less, uh, in their wake, uh, has said the same, uh, that it turns out there are two kinds of consciousness. Uh, there's the brainstem kind, uh, which we call wakefulness or the level of consciousness. Um, and then there's the cortical kind, uh, which we call awareness or the phenomenal, um, uh, contents of consciousness. So this distinction between a quantitative level contributed by the brainstem and qualitative contents contributed by the cortex, uh, is the way in which we save the cortical theory of consciousness, um, following Magoon and Maritzi's, uh, seminal observations. In case you're

wondering what this refers to, uh, it refers to a more recent study by Fisher and colleagues, which showed that the smallest lesion necessary to produce coma in human beings is a two cubic millimeter, uh, lesion. In other words, a lesion, the size of a match head, uh, in the parabrachial complex. So there's no doubt on the basis of Magoon and Maritzi studies all those years ago, and on the basis of these, uh, much more recent observations, uh, that the, that consciousness cannot exist without whatever it is that this part of the brain contributes this, this primitive brainstem core. Um, but this, uh, idea that it's merely a, a, a level or a, or a quantitative dimension of consciousness is what, uh, uh, I, I, I think, uh, is unsustainable. Uh, and I'm going to very quickly, because I'm sure many of you have heard me talking about this before. I'm going to very quickly whiz through the evidence for why this, this, uh, uh, notion that the brainstem contributes only a quantitative level, a sort of a wakeful prerequisite, uh, for consciousness, uh, rather than a quality and the content. Um, but I want to, before I do that, to just give you a clear picture of what is meant by this distinction. Um, the, the, the, the distinction, uh, is, is, is analogous to, uh, the difference between a television set and its power supply, uh, the idea that the cortex is responsible for the contents and qualities of consciousness for phenomenal consciousness, the real stuff, the real failure of consciousness, where this, this, the thing that all the arguments are about, um, it's that this is like the television set, and it has to be plugged in. Uh, of course you can't, a television set can't do its televisual thing unless it's powered up. Uh, but, uh, to claim therefore that the real source of television

and, and all that that word represents, uh, is the power supply, the electrical source, uh, in the wall, is, is ludicrous. Uh, and, and that's pretty much how we've all seen until very recently, how we've all seen what the brainstem contributes to consciousness. So nobody's doubting that it's necessary, that it's prerequisite, uh, that, that, uh, without it, you can't have cortical consciousness, but it's been, it's been, uh, uh, uh, reduced to a mere power supply, something which is not actually consciousness itself in the sense that, that, that we're all, um, arguing about. So why do I think that that's not correct? Well, uh, I'll start with the most dramatic evidence, which is patients who have no cortex. Um, if the theory that the cortex is necessary for the contents and qualities of consciousness, uh, then, uh, patients without cortex should be in a coma or at the very least, if we claim that the brainstem produces nothing other than the sort of background level, then patients with no cortex should have some kind of blank wakefulness.

Dr. Michael B. I suppose the closest, uh, natural kind of thing, uh, that tallies with blank wakefulness is the vegetative state, uh, which is also called non-responsive wakefulness. In other words, there's wakefulness without anybody being there, without it there being anything it is like, uh, for, uh, the, uh, organism that has blank wakefulness, uh, due to the absence of cortex. But when you look, um, at cases who, who have no cortex, and here's the scan of a girl, uh, with hydran and cephalo born with no cortex, uh, and you, uh, examine the girl herself, you see that she's not only conscious, in other words, not in a coma. She's conscious in the sense that she wakes up in the morning, goes to sleep at night. Uh, but more interestingly, uh, she is responsive. Remember my definition or the definition, uh, of the vegetative state, uh, the state of blank wakefulness, it is non-responsive wakefulness. And clearly this girl is not non-responsive. Here she's responding, uh, to the placement of her baby brother on her lap. And she responds emotionally in a way that any kid, any little girl, you place a baby on her lap. She'll, she'll, she'll most will express pleasure. Most, most kids like their little baby siblings being placed on their laps. Um, and it's not just her, it's most of these kids. Uh, if they're, if they're raised in something other than a state of complete deprivation, uh, like Romanian orphans, uh, who fall into what might look like a vegetative state, although their cortex is entirely intact. Uh, if you actually raise these kids in, in, in reasonably humane conditions, uh, they are conscious and responsive, uh, affectively responsive. That's the main point that I'm making these children with absolutely no, no cortex, their consciousness has a quality and the content, uh, that quality we call affect. Uh, it feels like something to be these kids. Um, and the content, uh, is not monotonous affect. It's all kinds of affect that he has a summary slide, uh, uh, extracting, uh, what beyond marker, who studied many, many, many of these kids, what he said about them, they show all of these different emotions. Um, and most importantly of all, they show them in situation, the appropriate ways. In other words, they respond with laughter. If they're tickled, uh, they respond, uh, with, with arch backs and, and, and, and, and protests, uh, if something that they want, uh, is denied them, they respond with a startle, uh, if there's a sudden, uh, unexpected noise and so on. So, uh, the evidence is that these kids are not only conscious in the sense of being awake, uh, which is interesting in itself. In fact, I can't imagine what a state of blank wakefulness might be like, but there you have it.

The, the, the, the, the, the concept, uh, derived from the green and Maritza's work is that there is such a theoretical fiction as blank wakefulness, but these kids are not blank. Uh, these kids are emotionally responsive, uh, in, in exactly the way that you would expect if they had consciousness, at least in the form of affect. So I'm arguing that, uh, with no cortex at all, these kids, uh, display the content and qualitative, conscious contents and qualities, uh, in the form of affects. Now, of course, the big methodological problem is that how do you know what it's like to be these kids? How do you know if there's anything it's like to be these kids? They can't tell us anything about what it's like to be them. So, uh, all you can do, um, in this situation, uh, is, is look at other lines of evidence. Uh, if we can find multiple converging lines of evidence, converging on the conclusion that the brainstem is where the affects are actually generated, uh, then, uh, we can do no better. We can do no more. We can do nothing other than that in science. So I want to show you now, uh, what happens if we look at cases who can speak, who've lost great swathes of cortex. Uh, we, we, we, they can't have lost all of the cortex because otherwise they wouldn't be able to speak to us. So I'm wanting to show you examples of cases who have lost great swathes of cortex and the cortex in question is the cortex that has been, uh, that, that, that has been most closely identified, uh, with, with, with sentient being with, with the subjective presence of a mind of a conscious mind. The first, uh, example is of prefrontal cortex. I think that it's fair to say that the mainstream view is that where all of this information processing comes together and is received by a sentient subject of the mind, uh, the mainstream view is that would be in the prefrontal cortex. Now there are few patients there. It's rare to have a patient with absolutely no prefrontal cortex, but here's one example. And I'm just giving you examples in, in, in this presentation. I'm, I'm, I'm speaking about individual cases, those individual hydranencephalic kids that you just saw, um, they are representative, uh, of the class of patient that I'm talking about. Here's a patient of my own, um, who has no prefrontal cortex, but happily, uh, has a sliver of language cortex. And so he's able to tell us what it's like to be him. Uh, I asked him if he was conscious of his thoughts and he said, yes, of course I am. And I explained to him that according to classical theory, he shouldn't be. So, uh, in order to confirm that he's conscious of his thoughts, I asked him if, uh, if he would be willing to solve a problem that would require him to consciously picture a situation in his mind. And he agreed to do that. So I said, I want him to picture in his mind. I want him to imagine that he has two dogs and one chicken, which he's, which he then says, okay, he'll do. And then I asked him, do you see them in your mind's eye? And he says, yes, he does. I then asked him to count the legs. You know, this is not an everyday task. It's something that's unexpected. So there he has these two dogs in his mind's eye and a chicken, and he has to count the legs. And of course the answer is 10. Uh, I asked him, how many legs do you see? And he says to my greatest disappointment, he says eight, I say eight. And he says, yes, the dogs eat the chicken, uh, which, um, I think is, uh, enough evidence for me that this chap, uh, is there, you know, that there's, there's, there's somebody at home, uh, who's looking at these dogs and the chicken, uh, and imagining perhaps concretely, you know, that the dogs actually at the chicken, uh, or perhaps, um, uh, he's, he's pulling a fast one. He's making a joke. Um, and in fact, the making of puerile jokes,

uh, is a very common feature, uh, of patients with the frontal lobe syndrome. And, uh, this is linked traditionally with the commonplace observation that these patients are, uh, emotionally disinhibited. Uh, and why I draw your attention to that is that, uh, if it were true that the cortex and the prefrontal cortex in particular is where affects are generated, then how come, uh, if you have an absence of prefrontal cortex, not only is affect preserved, but affect is excessive. Uh, there's, there's more affect than normal. There's, there's too much affect. Uh, this is entirely incompatible with the idea, uh, espoused by many, uh, respectable neuroscientists, uh, Joseph Ledoux, for example, who's a great, uh, authority, uh, on affect. Um, Joseph Ledoux says that of course, the amygdala and other subcortical machinery for anxiety doesn't actually generate any feeling at all, that the feeling of anxiety is read out to use his term labeled, uh, by the prefrontal cortex. And that's when it actually turns into a conscious experience. Uh, so I say again, if that were true, then how come, uh, not only my patient, uh, this one making this joke to us, uh, but all frontal lobe patients, um, the, the less frontal cortex they have, the more affect they have. Uh, so the less capacity they have to read out and label their feelings, somehow it seems as if they're beset by more feeling rather than less. Um, since I'm talking about feelings, uh, the other major candidate, uh, the other major cortical candidate, uh, as being the seat of the sentient self, um, is the insula. Bud Craig claims that the insula, or perhaps even just the anterior third of it, um, is where the sentient subject comes into being. And he links this particularly with affect. He says that this is the part of the cortex, uh, which registers, uh, the state of the own body. How do you feel now, uh, is registered by, uh, this part of the cortex. Uh, and it's on this basis that he claims that this is where the sentient subject comes into being. So again, just one case, who had absolutely no insula, bilaterally obliterated, and he interviewed in much the same way as I interviewed my patient with no frontal cortex. Um, and the Marsy asks him if he has a sense of self and he says, yes, he does. And I won't go through all of this, all of which is, uh, Demacia trying to, um, ascertain whether or not that this person is conscious and in particular, whether or not he possesses selfhood. Does he even understand what selfhood is? Um, and eventually Demacia says, what if I were to tell you that I know you better than you know yourself? Patience says, I would think you're wrong. Demacia says, what if I were to tell you that you're aware that I'm aware. Patience says, I'd say you're right. Demacia says, you're aware that I'm aware. Patience says, I am aware that you're aware that I'm aware. Please note all these references to I, uh, there is a sentient subject speaking to us here, uh, with absolutely no insular cortex. So I think the claim on the basis of, uh, the hydran and cephalic children, uh, the claim that the cortex is the seat of consciousness and of feeling in particular has to be dismissed, uh, but they can't speak. So we look at the other major candidates who can speak patients with no prefrontal cortex. They are conscious. There's a sentient subject there, and they feel like something they tell us and they show us, uh, and likewise patients with no insular cortex. Um, of course the claim can be made. Well, in these cases, uh, their consciousness arises from the remnants of the cortex, the remainder of the cortex, the quarter, the cortex that is not removed. Bear in mind, the ones who have no cortex, you know, uh, who clearly show, uh, uh, behavioral displays of

affect, uh, every prediction that you might make, uh, from the hypothesis that it feels like something to be, these kids is confirmed by their behavior, uh, by their responses to affective stimuli. Uh, and, uh, in that case we say, well, we don't know cause they can't speak. Uh, so then we use these other cases with no prefrontal cortex, no insular cortex who can speak. And then we say, well, it must be the, the, the other bits of the cortex that are generating the consciousness. It seems like, uh, losing wicked to me, but, uh, uh, let's look at other evidence than lesion evidence. Uh, remember, we're looking for converging lines of evidence, converging on the conclusion that the feelings are generated in the brainstem, in the brainstem. And so here we look at the case of a patient in, where an entirely different method is used. That is deep brain stimulation of a part of the reticular activating system. In this case, the substantia nigra, uh, the surgeon was aiming for the subthalamic nucleus, but the, the, the, the electrode went into the substantia nigra. And when, uh, he stimulated it, the patient fell into a profound depression, uh, within five seconds. Again, I'm not going to read all of this to you. It's, it will take more time, uh, than is necessary. Uh, uh, if you read it yourself, uh, you'll see this patient within five seconds felt fell into a profound depression. She has no psychiatric history, never been depressed before, never had a diagnosis of any kind, let alone a diagnosis of major depression. And here within five seconds, you can generate, produce, cause, a suicidal depressive state, uh, which within 30 seconds, uh, of switching off, uh, the, uh, the, the stimulant 90 seconds, I mean, uh, after the stimulation was stopped, the depression stops. Uh, very interestingly, uh, she, uh, agreed or very bravely, she agreed to have the stimulator, uh, switched on and off, uh, either in the subthalamic nucleus or in the substantia nigra of the reticular activating system, where she was blind as to where the stimulation was being applied. And every time it was applied in the reticular activating system, she fell into this severe depressive state again. So this is, you know, this is what you would expect. This is the, uh, the prediction, uh, from the hypothesis that ethics are actually generated in the brain stem, uh, uh, the, the, the, the prediction would be if you stimulate these reticular activating nuclei, you will generate affects. And that's exactly what you do. This is also why I mentioned the periaqueductal gray earlier, because, uh, if you stimulate reticular activating nuclei, you generate intense affects, but the same applies to periaqueductal gray. In fact, the greatest intensity and range of affects anywhere in the brain can be elicited from deep brain stimulation of the periaqueductal gray. Uh, brain stimulation of the cortical mantle produces very little, if any affect. Um, another method, positron emission tomography. So another study of Damasio's patients in states of, uh, uh, anger, uh, sadness, happiness, and fear. Uh, you see where the brain, uh, the, the, the, the major brain activity is in all of these affective states, the major activities subcortical it's in the brainstem it's not cortical. So again, uh, this is exactly what you would predict if the affect is actually being generated by the brain stem, not being read out by the cortex, but generated by the brain stem. Uh, and then there's also the matter of the neurotransmitters that psychopharmacologists tinker with, uh, the mainstream psychiatric medications, uh, modulate, uh, neurotransmitters that are sourced in the reticular activating system. Uh, we don't normally think of it that way, but it is simply the case, uh, that serotonin, which we increase, uh,

with antidepressants, uh, is sourced, uh, in the reticular activating system. Dopamine, which we block with antipsychotics, um, is sourced in the reticular activating system. Noradrenaline, which, uh, is blocked in many anti-anxiety drugs, uh, is sourced in the reticular activating system. If the reticular activating system was just the power source, uh, that I, uh, referred to in my analogy about the television set, then it should only interest anesthetists. You shouldn't be able to change the channel, uh, alter the programming of the television set by pulling out the plug or, or by, or by attenuating the power supply. Uh, you might expect the screen to sort of go blank or, or, or just flicker, but not to have whole new programs with new contents and new qualities. Um, and so this is my argument that the, the idea that feeling states affective feeling states of the kind that those children display, uh, of the kind that psychiatrists are, or, or, or directly trying to manipulate, uh, are generated by upper brainstem structures. They're not generated by cortex. You don't need to have cortex in order to have these states. And yet, uh, we all know and agree, we all agree that cortical consciousness is contingent upon reticular activating arousal. Uh, if we now are forced to accept that reticular activating arousal is not contentless, is not a mere quantity, is not just a level, uh, but rather has content and quality that we call affect. Then we have to accept that affect is a prerequisite for all other forms of consciousness. And it's on this basis, not on semantic one, that I'm claiming that affect is the foundational form of consciousness. Uh, it also makes good evolutionary sense, uh, that this ancient structure, uh, uh, rather than the cortical mantle, uh, which evolves only with mammals, um, that the ability to register your own subjective state, how are you doing effectively in terms of hunger and thirst and sleepiness, um, and, uh, to hotness and whatnot, um, that this should not require, uh, such a late evolutionary development. But that's another point. I'm just wanting to slot in there that I also mean it in the evolutionary sense. When I say that affect is the foundational form of consciousness, I don't only mean it in the functional sense. Now I'll pause there because that's a big claim and a very controversial claim. Uh, in fact, I must tell you, I'm still slightly dismayed that it remains controversial. Uh, I've been, I've been really surprised by the degree of pushback, uh, that I get, uh, when I argue and it's, I'm not the only one who argues that I'm not the first one to argue it. Um, but, uh, uh, Demacia argues it, uh, uh, uh, Panksepp was the first to argue it, um, Malka argues it, uh, and, uh, here I come a good few years on and argue the same thing, citing all of this evidence and, uh, wow, people just don't want to accept it. So let's hear what, what, what, what you have to say.

Stephen.

Stephen.

Yeah. Yeah. I'll read a question from the chat first and then we'll go to Stephen. So, Jonathan in the, in the chat wrote, I can imagine a modern large language model, the likes of GPT-3 creating such conversations without apparently having a mind's eye. While I agree, it is good evidence. I think we are seeing machine learning systems having similar conversations. So you've presented case studies that are the tip of the iceberg of broader populations and analyses. How do we think about our evaluation of conversational dialogue in questions about necessity and sufficiency here?

Stephen.

Yeah. You see, this is a kind of a, um, a relative of the philosophical zombie, uh, argument. Uh, the, the problem with this sort of thing, uh, is that, um, in fact, the same applies, uh, to you and me. I can't know for sure that Daniel is conscious, that Stephen is conscious, that David is conscious, or that the person who asked that question is conscious. You know, they, the, the, the, the, the point you've just made really does apply in exactly the same way. There's no greater reason to believe that it applies to the patients that I've just presented to you than, and that it applies to you and me, uh, because you too could be a philosophical zombie. In other words, you too could look as if, um, you have internal states, say that you have internal states, but actually there's nobody there. I can never disprove that. Uh, that is a metaphysical problem. That's not a scientific problem. Uh, the scientific method, uh, requires, uh, that your hypothesis has to be consistent with all the known facts. Um, and then you have to make falsifiable predictions, uh, from that hypothesis. And if your predictions are not falsified, then they are upheld, uh, unless, uh, and until, uh, they, uh, some evidence is, is found, which positively, this is what positivism is all about, which positively falsifies it. So I'm saying these kids look as if they're conscious, you know, they, they behave as if they're conscious. Now it's true that they might not be, and Stephen might not be, and Daniel might not be, and David might not be, but you know, um, my hypothesis is that they are. So what do I do? I say, well, if Daniel, uh, feels pain like I do, then when I stick a pin, uh, in his hand, uh, he's going to go, ouch, and withdraw his hand and avoid me in future. Uh, and that's, uh, you know, that's a confirmation of my prediction. Uh, the same applies with these kids. Uh, but you know, that's just one source of evidence. As I said, you don't accept just one, uh, you then say, okay, well, what about this line of evidence? What about that? And I've given you five lines of evidence here, um, in which in every instance, uh, the prediction from the hypothesis, uh, that these people are conscious and that the consciousness is generated in the brain stem and that the, that it has an affective quality, every single line of evidence, um, you know, the prediction is confirmed. What you would expect would happen if that were true is what happens. So, you know, I'm afraid you can do no more than that in science. Um, and it's a curious fact that we raise the bar, uh, of evidence, uh, when it comes to the problem of consciousness. Um, I think we must be very careful not to require, uh, any different sort of evidence in the case of consciousness than we require for any other, um, uh, uh, matter in science. Uh, and in this case, in any other biological function, you know, it's, uh, uh, uh, if, if you start with the assumption that consciousness is something special, um, that, you know, the ordinary rules of science don't apply to, then you're going to have a self-fulfilling prophecy. So yes, it's possible. It's possible that these kids and those patients are not there, uh, that they're lying to me, uh, that, you know, uh, it's, it's a little bit of a curious thing that somebody without a sense of self, uh, wants to pretend that they've got one. Um, I think you're on a kind of slippery slope there to, to, uh, to, uh, to irrationality, but I can't prevent you from holding that view. It's just not consistent with any of the evidence, I'm afraid. Thank you, Mark. Stephen?

Yeah, I was wondering how much you think the challenge with uptake is because people are sort of

stuck in the idea that knowledge and information is a signal coming in and meaning, it's all about meaning and the ability to communicate meaning as opposed to, um, engaging in meaningful action and skillful enacted processes of the voice twitching, the voice box twitching and enabling words to be in the space around us in some sort of way and whether that, that issue is, um, kind of a deeper ingrained challenge to overcome. Well, I completely agree with that. I think that that is the, uh, the, the, a fundamental source of the problem, um, because our common sense tells us our consciousness is extra receptive, as I said earlier, you know, the, our conscious experience is dominated by visual qualia and where do they come from? From, you know, things, light waves impinging on my retina from outside and I'm seeing things out there. And so consciousness seems, uh, to be so obviously derived from something out there. Um, and this is, uh, this was at the heart of the experiments that Magoon and Maruzzi did. Magoon and Maruzzi weren't expecting to find what they found. Um, they, they, they were expecting that, uh, a cat deprived of any extra receptive stimuli will fall asleep, um, if not fall into a coma. Um, and it was therefore a great surprise to find that that's not what happens, uh, that the consciousness is generated endogenously comes from deep within the brain doesn't flow in with sensory stimuli. Um, and, uh, when they lesion that core brainstem structure, the lights went out, uh, and, uh, they, they couldn't make sense of it. And they came up with this, with this, uh, theory, which as I've shown you, or at least as I've shown to anyone who's, you know, who's willing to look at the evidence and draw the conclusions, uh, that, uh, that, uh, that, that, that the, the, the, the way that they framed it is wrong. It doesn't explain what, what they saw. What they saw was that consciousness is endogenously generated. And where does that, where does the notion come from this common sense notion, um, that it flows in from outside? It comes not only from our own, uh, everyday experience, but also from the British empiricist philosophers who probably themselves derived it from common sense, you know, that the, the mind literally, the mind as a whole, you know, is, is derived from sensory experience, um, that leaves impressions, uh, leaves memory images impressed upon our cortex, uh, which are then associated with each other. And this, uh, according to British empiricist philosophy, which underpinned, uh, the, the whole of 19th century, uh, and early 20th century neurology and neuropsychology, uh, it's all predicated on that idea. So I think it's deeply ingrained within ourselves and within our science, um, and, and, uh, you know, within public, uh, perception generally, that consciousness comes from outside and, uh, you, you're speaking about some further, um, you know, ramifications of that and therefore sources of resistance against the alternative view, but all the evidence suggests that consciousness is endogenous, consciousness is generated from within. Um, and, uh, I mean, you saw, I, I have, I, I, I'm, I'm sorry, I don't know who the person was who made the comment about, you know, how do we know that this is not, you know, just some sort of program spewing out language. Um, you know, I have nothing against you. I don't know who you are. Uh, I just hear that all the time. Uh, and, uh, you know, I, I, I find it, frankly, I find it astonishing, you know, that you present all of this evidence and then still people say, well, you know, how do you know, how do you know that, that these patients are really conscious? And this is, uh, this, this,

this demonstrates the extent of the, of the prejudice, you know, that has come with that, with that historical, um, uh, with that legacy, uh, that, that I've, that I've, that I've mentioned. So yeah, I completely agree with you. Thank you. Steven with the raised hands, then Dave with the raised hand and we'll go with the raised hands. Okay. Thanks. Um, yeah. So sort of following on from that. Yeah. I totally agree. And actually I think the work with first, Carl Christian and active vision, um, is, is quite helpful. There's the idea that we're reaching out and feeling the world. Um, and I think actually I was looking back before 300, 400 years ago. Um, if you went back in time, people actually almost had that as a, an intuition of looking out in the world, but once we discovered the retina and then photography, we kind of re imagined. Um, so one question I've got is, is regarding cerebral palsy. I work a lot with communities with complex disability or non-verbal and their premotor cortex is often damaged. And in the 1970s, it was thought medically, if you couldn't speak, you couldn't think. So that was where they were then. If you couldn't communicate words. And then once they started using symbols, this symbolics and the ability to, and, and they suddenly discovered that all these clients who had, um, who were non-verbal actually could form complete sentences because they had a, a way to act in the world and interact in the world and develop, um, language syntax and all of that. So I, I'm just thinking whether you've had a lot of, uh, whether any of this works been tied in with clients with cerebral palsy. Um, not only with cerebral palsy, but with all sorts of, uh, neurological conditions. Now the, if I can just sort of piggyback on what you've said about cerebral palsy, the whole trend, um, within clinical neurology, um, has been us realizing that patients who we thought were not conscious actually are conscious. Um, the, the, and the whole re the, the, the sort of tenor of the research that's been done, uh, in recent decades, um, I mean, locked in syndrome of course is the extreme example, but, but I don't only mean locked in syndrome. I mean, all of these states of, of reduced and up-tundered, uh, content and clouded consciousness and so on, minimally conscious states, uh, uh, so-called vegetative states and so on. We finding more and more patients who actually, who we thought, uh, were not conscious with increasing technology. We find that they are, um, with, with improvements in our methods, we find that they are. So it's, it's a sort of an odd, um, it's an odd, um, contradiction with the sorts of philosophical skepticism and doubt that one confronts from people who are saying, well, you know, how do we know they're conscious that they, they, they, they're behaving as if they're conscious, but maybe they're not. Uh, in fact, what we've been finding in clinical neurology is patients who are behaving as if they're not conscious. Uh, and then when we, when, when we use the proper tools, we find that they actually are. So, um, uh, your comment about, uh, cerebral palsy, uh, can be multiplied your, uh, the observations that you refer to, uh, uh, consciousness, um, um, you know, using reportability, uh, as a criterion, the, the capacity to declare, uh, your conscious states, has really, it's, it's not just led to, uh, methodological and epistemological problems. It's really led to gross clinical errors. You know, it's ethical problems. Uh, I, I feel we need a lot more than mere philosophical skepticism, uh, before we can take seriously in neurology, which after all is a branch of medicine before we can take seriously, the possibility that these kids and

these patients look as if they're conscious, behave as if they're conscious, say that they're conscious.

And we say, well, we can't be sure. I mean, think about the implications.

Thanks Mark. Dave, with a question? Yes. Um, two, I'd like to say two things, uh, to what you were just talking about, uh, the difficulty of telling whether people who seem to be locked in or comatose are conscious. There have been some people who initially got their experience in working with the dying. They just wanted to people who were dying to not be alone. And they spent a lot of time as typically as volunteers sitting with folks who seem to be comatose, who seem to be completely unresponsive. And they found that events that unroll very, very slowly in the patient's room seem to elicit a response. And they got in high speed cameras and would, uh, record what the patient is doing. And that's a study that's also been done with, with infants, that infants seem to, to act extremely fast. Uh, and that if you speed up those images, you can actually see a child respond and who seemed to be just kind of lying there. They found with the comatose that it seemed they can carry out conversations and it actually did resemble conversations, taking a person through childhood scenes, for instance, and watching them respond and they nod or they'd shake or they'd grin. You mentioned some kid that used to bully you over a period of minutes, maybe saying it over and over or speaking very slowly. And you'd get this very slow, but eventually a full grin that a nurse would say, oh, that's just a grimace. Now, let me check her diaper. And the other thing is a much better, bigger picture question. Who actually is hyper alert? Getting out of professional settings or ideological settings. You walk around, you see someone who's just, you gotta be really careful with them. They will just startle. You have to be really careful what, what you say. You don't bring up unpleasant topics. You never imply that they've done something wrong. Who are these? For the most part,

these hyper alert individuals are children of alcoholics who as small children had to be the responsible person. They had to keep daddy from hurting mommy. Even. And I wonder how was Niels Bohr raised?

Was he hyper alert? Is that why he became so amazingly oppressive? As nice as he might have been. Nonetheless,

unremittingly insistent that only he was correct. So in that big picture thing, are the folks who want to insist that you absolutely positively are controlled only from the outside, were they the people who as children were never able to emit? They always had to react. They always had to be really, really careful because the environment really is something that will come in and get you.

Dr. Well, about your second comment, I will just say, please note how affective all of this stuff is that you're talking about. And I think that we really underestimate the importance of affect in terms of the whole gamut of phenomena that we call consciousness. But to your first point, I want to make clear that I am, I mean, I agree with what you're saying about the inference of consciousness in these patients.

Dr. But I want to make clear that I'm making actually quite a, in light of, or notwithstanding, or, you know, all of this other evidence and all of these other very interesting clinical phenomena, all of which point in the direction of, we've underestimated the presence of consciousness. You know, that we've been relying on very poor criteria in terms of reportability and so on, that really we need more

sophisticated methods. And the more we use more sophisticated methods, the more we find evidence that there is in fact somebody there. I agree with all of that. But I'm making a very, very limited claim. Dr. I'm saying, I'm not saying these kids are like you and me. I'm not saying they have a full rich inner mental life. I'm saying only that there is something it is like to be them. And in light of all the evidence in their cases, plus all the other evidence that I've cited, or using all these different methods, all I'm saying is that that something it is likeness that these kids possess, is it feels like something. They have feelings. That's all I'm saying. So it's all very interesting, all these other things about how all these other capacities have been missed by us in relation to all of these other neurological conditions. But in these cases, which have no cortex whatsoever, they have only some of them have basal ganglia and, you know, other subcortical forebrain nuclei. Some of them don't. But even in cases you have no forebrain, only a brain stem, they display affects. And so I'm saying that the brain stem prerequisite form of consciousness does have a quality and the content, therefore feels like something, there is something it is like to be these kids, and that quality and content is effective. That's all I'm claiming. That the foundational form of consciousness is affect and it's generated in the brain stem. So it's actually, I think, a rather modest claim in light of all the evidence. It's quite a conservative claim. It's a sort of a deflationary claim. But so much flows from just that. And I'll just say once more, I find it very hard to understand. I mean, I can understand it psychodynamically, you know, or politically, or what, you know, I mean, people's whole careers, I mean, generations of careers have been built on the assumption, you know, that the cortex and consciousness are intrinsically bound up with each other, which, you know, so the people have a hell of a lot to lose. And and many of us with egg on our faces, and I freely admit that I, too, believed that the cortex was the organ of consciousness. This is what I took in like mother's milk when I trained. And I believed it for many, many years after I trained, you know, and so it's, you know, I'm not exempted from that. But gosh, we've got to look at the evidence. It's wrong, we made a mistake. You know, that doesn't mean the cortex does nothing. It doesn't mean it doesn't contribute anything to consciousness. It's just not the source of consciousness. The cortex is elaborates this very raw primary elemental feeling that is generated in the brainstem. And all we need to do is accept that. And then we'll make huge progress in consciousness science. The fallacy that the cortex is the source of consciousness is really a serious impediment to us at this point in the history of our field.

Daniel. I think it's you next, Daniel.

Daniel. Yes, it is indeed a theme of the discussion that ambiguous evidence are continually interpreted and reinterpreted in light of our inherited or our previously structured priors. And that's true in the active vision example. It's also true in the scientific example, where our inherited priors about scientific models influence how new evidence is not just perceived, but funded and carried out. And it made me think about when people activate a gene or remove one gene and make claims of function might be an analogous to the brain, but removing or having one node in a complex network showing that there is a single point of failure or gain of function does not entail that as a generating mechanism, which is just an example of an explanation that retains

an alternative perspective. And so I think it's interesting on multiple levels as we kind of apply this idea of updating and learning and structural learning on our theories about ostensibly the part of the body that does this function as opposed to one's hand, which is important and has nerves in it, but doesn't do this kind of engagement.

Dr. Thanks very much. So I agree. That's an everyday observation that we are reluctant to accept prediction errors. We cling to our priors, especially our most deeply consolidated priors.

And we see this in everyday life. Nobody likes to find they were wrong, that they have to change their mind, that they have to accept evidence for review other than the one that they had always held. And this applies even to us scientists. I mean, who in their graduate studies, when they were doing their first research, didn't hope that their hypothesis was right. And then when the evidence came in and they found, oh dear, I haven't found what I expected to find, it's not welcome. So what I'm referring to, based on what you've just said, I know this is not quite what you were saying, I'm sort of elaborating what you were saying, is that we don't recognize sufficiently the affective dimension in our resistance to prediction errors. We all know about priors and posteriors and how this all works mechanistically. But just from the sort of anecdotes that I've mentioned, I'm trying to get us to recognize that there is an affective dimension to that. And when we get to this point about the concentric predictive hierarchy, I'll come back to that. But I think it's an interesting and important observation that as you traverse the hierarchy and get to what, in my way of thinking about it as a concentric hierarchy, to get deeper into the hierarchy, in the traditional way we speak of going higher in the hierarchy. But as we go deeper, further from the sensory motor periphery, so we come upon priors which are more and more difficult to update. And I say again, there's a feeling that goes with that, which is, I don't care what you tell me, I'm not going to believe you.

Thank you, Mark. Stephen, with a question? Thank you.

Yeah, I know you're going to come back to the cognitive consciousness, but just something around terminology,

we've been doing work around an ontology working group at the Active Inference Lab, actually, trying to understand how to think about the language.

And you mentioned it could be affect is generated in the brain stem, and then, and that's when something's like something,

that's what it's like. But I'm also wondering how you take in the dynamic with the body and environment going down.

I don't know if that's seen as another manifestation, because obviously the brain stem makes sense, because it's more integrated with the rest of the body's nervous system, it's less distant from it.

But I'm wondering whether the term generated within is more to do with when it's connected to the rest of the brain,

and when it goes down, it's actually more distributed in the mind body environment dynamic.

I would have nothing against that way of thinking. I'm very sympathetic with it.

I'm not a narrow localizationist. I'm very, very happy to recognize.

It comes naturally to me to think about functional systems which are distributed and dynamic.

And in fact, the reticular activating system is probably the most deeply interconnected with everything else in the brain system that there is.

It has tentacles everywhere. And not only, although it's called the ascending reticular activating system, it descends also.

It modulates the whole nervous system in very important ways that that will come to, because this whole thing of precision modulation is at the heart of my way of understanding consciousness. So the main thing that I'm wanting to say is I agree with you that this is a deeply interconnected system which functions by modulating other systems.

And I would be very happy if we had to leave it there and say, well, you can't quite say that affect is generated in the reticular.

In the reticular activating system because, you know, it's, it's really modulating these other systems.

My reluctance, however, in this case to accept that is twofold.

The one is that that's the kind of thing that people like Joe Ledoux are saying.

They're saying, yeah, this is all unconscious until it is labeled by or read out in cortex.

You're not talking about cortex. So, you know, that's, but the principle remains, you know, that I don't want to concede the point that it only becomes conscious once it modulates something else.

And that that's really where the contents and qualities are.

That's a sort of theoretical reluctance on my part.

You can see why.

But more importantly, more telling me is the evidence.

When you look at these kids, they have no cortex.

So their affects can't be modulating.

It can't be the result of something that the brainstem is doing to cortex because there ain't no cortex.

And yet, you know, there's all the evidence that they have affects.

And so I think as much as surprising as it may be, you know, it's not consistent with our normal way of thinking about cortex.

And it's also not something that I'm wedded to the idea that, you know, anything is generated in anything.

But the evidence suggests that certainly the modulatory influence that the reticular activating system has over the cortex, true as it is important as it is fundamental as it is to how that system works.

That the affects are not a product of the modulation of cortex.

The affects are generated in the subcortical structures.

And if you just take a step back, the implication of that is that, well, maybe there are affects in vertebrates who don't have cortex.

In other words, maybe affects are not the exclusive preserve of mammals.

And actually, that seems pretty likely.

It seems pretty darn likely that non-mammalian vertebrates have affects.

And again, you know, when you start looking at the evidence, it's pretty darn good that they do.

So I'm happy to accept everything that you're saying about let us not forget that this is a system which is deeply interconnected with other systems,

that its primary function is to modulate other systems.

But all I'm saying is that when it comes to cortex, it's not that the affects themselves are not generated by the modulating of cortex.

That doesn't mean that there's a disembodied brain or disembodied reticular activating system, but you can have a decorticated reticular activating system, and it can generate affects nonetheless.

Thank you, Mark. Dave, do you want to go?

Thanks.

Okay. So one thought.

I see that the comments in the chat line have been dealt with. Can I move on?

Yes. Let's try to get to at least one other of these important points.

Oh, well, this is a brief point.

You know, the reason I'm making this point about affect not being synonymous with interceptive inference is because those few people working within the active inference framework who are not corticocentric, and happily they're increasing. You know, there's people who worked within this framework, they had no reason to question, you know, all of the mainstream assumptions of cognitive neuroscience as far as cortex is concerned.

But slowly it's become apparent that affect is an important topic that we're not only talking about perception and action and cognition,

you know, we're talking also about affective processes and about body regulating processes and so on.

In fact, the whole of self-organization has to do with the regulation of the own body and the whole notion of self-evidencing systems and so on.

So this is happily slowly coming and increasingly coming into the mainstream of the people working within the sort of active inference framework.

But the dominant view within that school of thought is that affect is to the internal body what perception is to the external world.

In other words, extra perception is perceptual inference regarding things going on outside of me and affect is perceptual inference about things going on inside of me.

And that's, you know, if that's, you know, if that were so, that would be nice and simple.

But I don't see it that way.

So, and perhaps the outstanding proponent of the view that affect is interceptive inference is Neil Set.

And he's just brought out a really very good book.

I have it right here.

Here it is.

Being You, you know, which is predicated on that view.

So because it's because I agree with him about so many things, I think that it's easy for people to conflate our views and to imagine that what I'm saying is the same as what he's saying on this school.

And why do I think that affect is not the same as perception, but rather interceptive perception?

Well, first of all, because there are many interceptive perceptual states which are not affective.

So, you know, you can perceive your heart rate, the beating of your heart.

You can perceive your respiration rate or the inspiring and expiring of air from your lungs.

You can perceive gurgling in your tummy, which is an unequivocally visceral state.

And all of these things, you can perceive them indifferently.

You can perceive them without any affective valence or quality at all.

So if affect were synonymous with interceptive inference, then why are these common and garden examples of interception not affective?

And then secondly, if affects were interceptive perception, how come so many affects are generated by extraceptive events where there doesn't seem to be any clear interceptive driving force.

So, for example, pain, which is a fairly simple example of an affect.

You know, you stick a pin in my finger, it comes from the outside and it penetrates the nociceptors in my extra receptive nervous system.

And I feel this intense affect called pain.

So that's extrareception and yet it's affective.

The same applies to surprise, you know, to fright.

You know, you get a shock.

Well, in fact, electric shock is another one.

But these are, but I mean just, you know, fright.

Something unexpected happens outside of you and you startle.

And there's an affect that goes with that.

A particular type of disgust is another one.

You imbibe something that tastes or smells bad and you wretched up with this terrible feeling of disgust.

And then they're more, and they're not purely entirely non bodily affective states like missing somebody, you know, missing your brother.

It's clearly affective and there's nothing bodily about it.

So, and nothing intercepted about it.

When I say nothing bodily, I mean, particularly, you know, it's not, it's not about the state of your interceptive visceral processes.

So I'm saying there are interoceptive visceral processes which we perceive which are not affective.

And there are affective processes, affective states that we experience which are not interceptive.

So I don't think that we can say that affect is interceptive inference, that it's the same as external perception, except it's directed toward the state, the internal state of the own body.

I think that it's something else.

And I don't say this to be difficult.

You know, I'm actually quite concerned that those few of us within the active inference framework who are recognizing the importance of affect,

can't agree on this point because, you know, there are few enough of us, you know, who recognize the importance of affect.

So I'm just wanting to make clear why I think that it's not merely a kind of mirror image reflection of of extraceptive influence.

And then I want to add one last point, which is that when you read, I mean, everything that I'm going to go on to argue about why extraceptive perception.

Remember what I said at the very beginning about cortical vision, having been the model example of where we will find the neural correlative consciousness,

which has led us down this blind alley and got us into such a mess.

That it has to do largely, as I'm going to argue later, with the fact that cortical processes and perceptual processes are not intrinsically conscious things.

You can perceive the world through all your extraceptive modalities, vision included.

At the cortical level too, you can perceive the world without perceiving it consciously.

In other words, extraceptive inference, perceptual inference is not an intrinsically conscious process.

There's nothing about extraceptive inference or perceptual inference that requires consciousness for us to understand the mechanisms at work.

And so if interceptive inference is just the same thing applied to the interior of the organism, it suffers all the same problems.

That it's not clear why it becomes conscious.

And I said that's going to be my last point on the score.

But in fact, I realized there's one more that I want to make, which is that the vast, vast, vast majority of our interceptive processes are unconscious.

I mean, even more so than our extraceptive processes.

Most of our regulation of our visceral, bodily economy happens autonomically.

We know nothing about it.

So it would be, again, a very curious place to find the essential mechanism of consciousness, and in this case of affect, if the majority of the mechanisms of this kind, in fact, do not operate consciously.

And if it suffers the very same problems as our extraceptive, using vision as a model example, using any form of perception, I think, as a model example of consciousness is to make a mistake.

Because consciousness is not an afferent process.

Consciousness is an endogenous process.

It's something generated not from within the guts, but from within the core of the brain.

And that's where we're going next.

So I'll stop there for any comments on that point before I move to the next one.

Great presentation, important distinctions and points.

So indeed, we'll take a breath.

And I think we can head on to the next one.

Okay, thanks.

So now, I've told you in relation to the point I've just made, I think that the hard problem of consciousness has arisen in large part from the example that we've used of consciousness.

So I'm starting with Crick, who has to be credited with bringing consciousness into center stage.

So, you know, I'm not bashing Crick.

And I also have said already, I think it's perfectly reasonable that he started where he did.

You know, it's clear why he started with visual perception as his example, where we should look for the neural correlate of visual consciousness, and then distinguish, you know, unconscious vision from conscious vision.

And then there we have it, then we have the neural correlate, and then we can generalize from there.

I'm saying that it turns out that was an unfortunate place to start.

And it links with what I was saying earlier.

That's also why I'm saying, I don't want us to make the same mistake, just turning it inwards and saying we're dealing with perceptual processes.

So let me say why all of this casts new light on the hard problem.

Yeah, so he asked Chalmers, the person who formulated the hard problem in the very paper that he formulated it.

And he says, what makes the hard problem hard and almost unique is that it goes beyond problems about the performance of functions.

To see this note that even when we've explained the performance of all the cognitive and behavioral functions in the vicinity of experience, there may still remain a further unanswered question.

Why is the performance of these functions accompanied by experience?

A simple explanation of the functions leaves this question open.

Why doesn't all this information processing go on in the dark, free of any inner field?

So this is a core aspect of Chalmers' hard problem.

He's saying that our normal way of going about doing cognitive neuroscience, where we take a mental phenomenon and we say, OK, let's see which part of the brain needs to be damaged in order to disrupt this mental state.

And then, you know, let's start piecing together what this part of the brain does and how it works.

And next thing we end up with a wonderfully complicated map of all of the mechanisms that work in this case, using the standard example, visual perception.

Chalmers' point is that you can do this.

This is the standard approach in cognitive neuroscience, but it doesn't tell you why there is something it is like to see.

So you can, you know, the famous knowledge argument with Mary, the visual neuroscientist, she knows everything that there is to know about the functional mechanisms of visual perception, but she knows nothing about what it's like to see.

In the example that Jackson or the story as Jackson told it, you know, she's never seen color.

But, you know, I like to extend and simplify his arguments and just say she was born blind.

She knows everything there is to know about the functional mechanisms of vision, but she's never experienced vision.

And if one day she's blessed with the gift of sight, she will learn something utterly new about vision, namely what it is like to see what redness is like, what blueness is like and so on.

And this is their argument. The argument is that the normal way we proceed in cognitive neuroscience,

namely to dissect the function and in this way to explain the mental thing that we're trying to explain, works very well for all of these mental functions.

But it doesn't work very well. It doesn't work at all for consciousness. Consciousness defined as what it is like to perform these functions. So they claiming Chalmers is and Nagel and others of that ilk.

And so I think that somehow that somehow the consciousness is in a different realm from every other mental function, because, you know, when you when you've explained all the functional mechanisms, you still are left with this epistemic gap between that explanation of the function and the qualitative experiential phenomenal feel of seeing.

And so I'm saying that the problem is that they're using perception as an example.

Perception is not an intrinsically conscious function. And indeed, cortical perception is not intrinsically conscious.

That it is demonstrably the case that the cortex can perform uniquely cortical perceptual operations like actually recognizing color.

You can recognize color unconsciously. You can recognize faces unconsciously. You can read words unconsciously.

So if the cortex can do all of this perceptual and cognitive gymnastics without consciousness, I think it's perfectly reasonable to say, well, why doesn't all this information processing go on in the dark without any inner field?

Because that kind of information processing can go on in the dark without any inner field.

Perceptual inference is not intrinsically conscious.

So this statement of Chalmers is this really strong statement where he says there is no cognitive function such that we can say in advance that explanation of that function will automatically explain experience.

This is what he says on the basis of everything that I've just told you. This is where he's led. He's saying that explaining the function can never explain experience.

If that's the case, we're in big trouble because consciousness is somehow going to have to be dealt with in a different way from everything else in science.

But if you change that word cognitive to affective, would Chalmers have said this?

There is no affective function such that we can say in advance that explanation of that function will automatically explain experience.

Why it feels like something because that is the function of feeling.

The function of feeling is for the organism to feel it, to feel its own state.

So I'm saying had we begun there, had we begun our quest for the neural correlates of consciousness with this model example, had we begun by asking why and how does it come about that feelings are not experienced, I think that we would have experienced, I think that we would have had a much less hard time of it than we would have had a much less time.

We have had with starting with visual information processing.

The mechanism of vision, the mechanism of vision, the mechanism even of cortical vision is not intrinsically conscious.

It does not automatically explain experience because it does not automatically have to be experienced.

But the mechanism of feeling, you cannot explain how feeling works, why feeling exists, what it's there for.

If you have a functional account of feeling, it would have to account for why it feels like something.

In other words, why it's experienced.

So I think that what I said in my earlier point that these facts cast new light on the hard problem.

This is the new light that it casts.

I just have also on the screen this slide, this famous review paper by Kilstrome, which summarizes all the experimental evidence for the conclusion that we were led to actually a good four decades ago already.

That perception without awareness of what is perceived is perfectly commonplace and learning without awareness of what is learned is perfectly commonplace.

Cognitive and perceptual functions performed by the cortex are not intrinsically conscious processes.

So this question of Chalmers is why is the performance of these functions accompanied by experience?

Why doesn't all this information processing go on in the dark?

That perplexity is not difficult to understand in light of those facts.

But I don't think that this question has anything like the same gravity when you apply it to the kinds of mechanisms that give rise to feeling.

I'm saying in short that if Mary was an affective neuroscientist rather than a cognitive neuroscientist or a visual neuroscientist, she could not know everything that there is to know about the functional mechanisms underpinning feeling without expecting that it will generate something that is experienced without the mechanism necessarily accounting for why it is experienced.

And this is why I think shifting our focus from cortex to brainstem and from perception to affect has big implications for our ability to finally come to grips with this hard problem of consciousness.

I'll pose there again.

Thank you.

Thank you, Mark.

Thank you, Mark.

I'll read a comment in the chat and then you can respond then Steven.

So Dean writes pre arousal plus awareness minimum of two, not as brain geography or territory establishment effort, rather a dance in an evolving relationship as we might see in an agent dancing with the environment.

Whereas the territorial view might examine control brainstem of the cortex or vice versa, one in a controlling relationship through a mechanical lens.

This relationship of arousal and awareness could be choreographed, but not choreographed mechanized, but not necessarily choreographed.

Not sure what that part meant.

Are feelings a residue of arousal plus awareness revealed?

We accept the signals like we dance out a path generated.

Look, I think that this is similar to the point that we were discussing earlier, you know, about generated in, you know, the upper brainstem as opposed to the upper brainstem modulating other structures, whether they be above or below the brainstem in the nervous system.

So I said, then I'm very happy to think of it as a dance, as something choreographed where it's not a matter of who's master and who's servant, you know, that's a, it's a, when it comes to the relationship between

affective arousal and cortical representation.

I think that there's some instances where affective arousal just absolutely brushes aside any amount of decision making and thought processing.

And conversely, the opposite happens too.

You know, that there clearly are ways in which one can, in a top down way, modulate affect dramatically.

And I don't only mean, you know, regulate your emotions.

I mean, if you look at some of these examples of these Buddhist monks who are able to, you know, really massively control autonomic functions and the affects that normally would go, you know, with not breathing or walking on, you know, painful surfaces and so on.

Those affects seem to be overridden.

So nothing against the idea of there being a choreography between brainstem and cortex.

But I've already said why I think that in this instance, the affects do not require the cortex to be actually qualify for the name affect that they actually are felt without cortex.

I've already said why, what the evidence is, both in terms of studies of, you know, here and now studies using the various methods that I mentioned.

The most dramatic being, as I said, cases where there is no cortex, and yet there's evidence that there's plenty of affect there.

But also the evolutionary implications of that, which is that we don't have to wait for the appearance of cortex and the mammalian series for the dawn of consciousness.

But that if affect is generated in the subcortical structures, then affect does not require cortex and affect is, it can be, can be attributed to other vertebrates.

And not only vertebrates, by the way, there also is good evidence that some invertebrates are conscious.

The octopus is the famous, the most famous example.

But I think that there's a further problem in this comment, you know, which has to do with the word awareness.

You know, you speak of arousal and awareness, and you seem to attribute arousal to reticular activating system, which I'm happy with, and awareness to cortex, which I'm not so happy with, if you mean that you cannot be aware of arousal without cortex.

So I think this word awareness, it's like all the other words I mentioned at the beginning, like emotion and affect and so on.

And I said, I want to, I want to use the word feeling to make clear what I'm talking about, because these words mean different things to different people.

The word awareness means different things to different people.

And to, to, to many people, it seems to mean something like reflective awareness.

In other words, you know, in other words, you know, in other words, being aware of what you're aware of, or access consciousness, as Ned Block calls it, you know, that there's this sort of phenomenal consciousness, which is some kind of like a raw consciousness.

And then there's a, then there's a subject that, that accesses this information.

And it's, it's like the global workspace theory works that, I mean, it's pretty similar sort of idea.

If, if, if we're going to reserve the word awareness for that higher order thought or reflective consciousness or

access consciousness, then it means that we can't attribute awareness to raw fields.

And, you know, that's, that's an odd use of the word awareness to me.

To me, how can you feel pain without being aware of it, you know, and, and pain, you don't need to have any kind of reflective cognition about it.

You don't need to have any explanatory labeling or, or anything.

It's just like, shed that saw and you pull away, even if you don't have language, even if you're not human, even if you're not a mammal, even if you're another kind of creature.

So I arousal for me implies awareness in the way I use the word.

But if, if, if we're going to use awareness for that higher order type of consciousness, then I would just want to remind you that there are other types of consciousness that don't possess that kind of awareness, which nevertheless have something it is likeness.

And, and, and the core of that quality is feeling that something it is likeness is feeling.

Thank you, Mark.

Stephen.

Yes, thank you.

So I was wondering if we think of consciousness as a kind of an integration of our action expectations of what's happening.

And those expectations then will, and Caspar Hesp talks about this with his work with active inference, is how those expectations were, weren't met and be effective.

Um, so I was wondering within that kind of, that prediction process, um, when someone gets a question, right, say a mathematician, when they get it right, they get a feeling that it's right.

It's like they, it's a felt sense.

And you could ask someone what's that feeling like, and they can sort of bring out a metaphor landscape that relates to that.

Like it can actually have a structure.

And there's a similar thing with pain as well.

Pain has to be expressed as a metaphor.

You have to say it's like a stabbing or it's a, a throbbing or it's a, it's always some kind of dynamical kind of metaphorical embodiment.

Um, so I'm just curious, um, as these, you know, this embodiment, this meeting in this brainstem, uh, how you, how you, how you see that in terms of if that's congruent with you and whether this, this is then a sort of a re, a re-imagining consciousness, like a re-imagining of these, of these expectations.

Um, and it can, it could go up or down, um, in terms of how that is being kind of, um, realized.

So I'd, I'd like to separate out, uh, two different aspects of what you're saying.

Um, the, the first has to do with metaphor.

Um, and, um, I think that, that may be true of, um, of being able to report or declare your conscious states in language, uh, that somehow or other you have to symbolize because language is, is just that.

It's a symbolic representation, um, of the, uh, thing itself.

Um, so, uh, that, that, that, that, as long as you, you, you use the word expressing it, uh, yes, you can't express, um, in words and emotion without using some sort of metaphor, some sort of symbolic system, uh,

in order to be able to communicate it to others.

Uh, the, the, the, the, the, the, the, the, the, the can of course be the direct behavioral affect of display expressing of the emotions in Darwin sense of the use of that, uh, uh, that which we then, um, us, us symbolic species, um, you know, we, we, we, we use that symbolically, uh, too.

But I don't think that the raw feeling itself has to be re-represented in any way.

I think it can just be the raw feeling without it being represented in any language in the broadest sense of the word.

And so I don't think that metaphor is a necessary component of the core feature of feeling.

Then the other thing you were saying, which had to do with surprise, if I understood you correctly, I, at times in my own thinking about this, there were times when I was keen on thinking about affect or linking affect more with error signals than with inferences themselves.

And that was, you know, I was thinking along those lines because so many people within the active inference framework were speaking about perception.

You know, this famous phrase, controlled hallucination, that perception is fantasy, that it is the upshot of the predictive process.

And what worried me about that is the very thing I've been banging on about all along, you know, which is that perception is not intrinsically conscious.

And so this business of inferring the source of your sensory signals doesn't explain why there's something it is like to infer the source of your, the cause of your sensory signals.

And so, so, so from that sort of point of view, I've, I found myself thinking, well, it, it's not so much that, you know, I mean, imagine if what you perceive is just what you predict, that that, that would be, you know, bloody dangerous.

So the controlled aspect seemed to me to be the more important aspect.

It's the, it's the constraint on what you predict.

That seems to be more the upshot of what we call perception as, as opposed to hallucination.

And so I wanted to place the emphasis more on, on the error side of things than on the, than on the prediction side of things.

But subsequently I, I have changed my mind on, on, on that score for, which will become clearer.

You know, when I, when I get down here, where is it?

When I get down here, which we obviously not going to get to today, but I'm, I'm, I'm very much looking forward to us having a second meeting where we can get there.

For me, it's, it's got everything to do with the, with the modulation of confidence in the prediction versus the error signal or the error signal that is, that, that is connected with that particular prediction.

And, and, and, and I will just say it now because we're not going to get to it in this meeting, that, that the pithy formulation of the way that I see it is that it is good when things turn out as expected.

And it is bad when uncertainty prevails.

And I mean, yeah, goodness and badness, um, in, as, uh, in the, in the valuative sense of the word that is linked to the primary, um, value system of all living things, you know, which is that it is bad to die and it is good to survive.

And if things are turning out as expected, and if things are turning out as expected, in other words, you have a prediction, uh, which actually, uh, cuts it, uh, that, that is good, uh, from the viewpoint of the organism.

Um, and that if things are not turning out as expected, uh, and uncertainty is, uh, prevailing, that that is bad.

It, uh, it, it, it means, you know, this, this is likely to end in tears.

And so, um, the, the, the, the, uh, what was it?

The choreography that somebody with the dance that somebody else was talking about earlier.

And I think that what, uh, that, that, that, the, the crucial thing here, uh, is the, is the, is the palpating of the precision, uh, the modulating of the precision, uh, in the predictions.

Uh, uh, uh, over the error signals, uh, and that, that is at the heart of what affects all about.

And I'm only saying that now, uh, in advance, because I know we're not going to get there, um, in, in, in this meeting because we've only got seven minutes to go.

Uh, but I'm going to try if I may, unless there are any further questions and comments, are there any further questions and comments?

No, here are some future comments.

It takes many.

So then if there are still something, I think what we should do is I'll take, I'll take whatever comments remain or questions, and then we'll start next time with this point, because I don't think I can squeeze it in.

Uh, to, I don't think I should even try to squeeze it into, to, to shorter time period.

So, so, so yeah, that was my comment was, uh, we will have a last few minutes to, um, really just appreciate you coming on for the 16.1.

And we'll find a time to do 16.2 when we will condense and sort of reflect on what has already transpired and look forward to these next points.

I think it was really great to walk through it this way.

So Stephen or Dave, do you have any last thoughts?

Otherwise, Mark, please have a last thought.

So Dave first, then optionally Stephen and then Mark.

Yeah.

Yeah.

In addition to Professor Seth, whose book we already have, who else can we, uh, check with as people doing AI without the, um, the, the, the constriction?

With, without which you mean, with, you mean who else is working on affective processes within, within this?

Exactly.

Yeah.

So, um, two people who come to mind, uh, immediately, uh, is Manos Tsakiris, Manos Tsakiris, uh, and Katerina Fotopoulou.

Katerina Fotopoulou.

Uh, they, they, uh, uh, they both great.

Um, and, uh, uh, and also both of them are very good speakers.

Uh, so, so if you, if you're wanting to invite them onto, uh, this, this, um, seminar, what, what live stream

series, I think that they would be, um, they would do well.

Awesome.

Thank you very much, Stephen.

Yeah.

Just to say, thanks very much.

And, um, yeah, we, we've been reading some of the work of Ryan Smith and Caspar Hess.

So, oh, well, Ryan Smith also, uh, Ryan and, and Maxwell Ramstead.

Uh, they're also working on affect.

They both, they, I should have mentioned them, Ryan Smith and, and Maxwell Ramstead.

Sorry.

Uh, they, they're doing some fabulous work.

I really loved their recent paper on folk psychology.

Uh, it, it was just brilliant.

Excellent.

Oh, thanks.

Well, we'll, we'll, we'll be connecting those dots up.

I'm sure next time as well.

Thank you.

Okay.

Well, um, thoughts.

Yes.

Yeah.

Just, just to say that I appreciate having the, um, I mean, it's not every day that you have two hours, uh, with, with, with excellent minds, uh, to be able to talk about only half of what you wanted to talk about.

So it's been very kind of you, uh, to, uh, to, um, give me this, this, this latitude.

Uh, I think that these issues, and I'm also particularly pleased that you're going to be in one way or another, uh, uh, recording.

And, uh, I don't just mean video recording, but also turning into some sort of written form, uh, what we're talking about, because as I hope, uh, you can see from what I've said, just in half, uh, of what I wanted to go through.

These really are pretty basic issues for, for consciousness science and for the science of the mind and for, and, uh, and, and by implication, uh, for what we can do within this active inference framework, uh, uh, uh, in, in tackling these.

Really, I don't mean to be hyperbolic when I say these profound and ancient problems.

I think that we have real prospects of being able to make significant progress, um, on, on, on these fronts.

And so it deserves is the point I'm making.

It deserves the time that, that we are giving it.

And I'm grateful to you for, um, for recognizing that.

Well, sir, thank you very much.

Thanks very much.

See everybody next time.

Thank you.

Thank you.

Cheers for now.